

## **Open Shortest Path First (OSPF)**

Open Shortest Path First (OSPF) is a routing protocol for Internet Protocol (IP) networks. It uses a link state routing (LSR) algorithm and falls into the group of interior gateway protocols (IGPs), operating within a single autonomous system (AS). It is defined as OSPF Version 2 in RFC 2328 (1998) for IPv4. The updates for IPv6 are specified as OSPF Version 3 in RFC 5340 (2008). OSPF supports the Classless Inter-Domain Routing (CIDR) addressing model.

OSPF is a widely used IGP in large enterprise networks.

### **OSPF messages**

Unlike other routing protocols, OSPF does not carry data via a transport protocol, such as the User Datagram Protocol (UDP) or the Transmission Control Protocol (TCP). Instead, OSPF forms IP datagrams directly, packaging them using protocol number 89 for the IP Protocol field. OSPF defines five different message types, for various types of communication:

#### **Hello**

*Hello* messages are used as a form of greeting, to allow a router to discover other adjacent routers on its local links and networks. The messages establish relationships between neighboring devices (called adjacencies) and communicate key parameters about how OSPF is to be used in the autonomous system or area. During normal operation, routers send hello messages to their neighbors at regular intervals (the *hello interval*); if a router stops receiving hello messages from a neighbor, after a set period (the *dead interval*) the router will assume the neighbor has gone down.

#### **Database Description (DBD)**

*Database description* messages contain descriptions of the topology of the autonomous system or area. They convey the contents of the link-state database (LSDB) for the area from one router to another. Communicating a large LSDB may require several messages to be sent by having the sending device designated as a master device and sending messages in sequence, with the slave (recipient of the LSDB information) responding with acknowledgements.

#### **Link State Request (LSR)**

*Link state request* messages are used by one router to request updated information about a portion of the LSDB from another router. The message specifies the link(s) for which the requesting device wants more current information.

#### **Link State Update (LSU)**

*Link state update* messages contain updated information about the state of certain links on the LSDB. They are sent in response to a Link State Request message, and also broadcast or multicast by routers on a regular basis. Their contents are used to update the information in the LSDBs of routers that receive them.

## Link State Acknowledgment (LSAck)

*Link state acknowledgement* messages provide reliability to the link-state exchange process, by explicitly acknowledging receipt of a Link State Update message.

## OSPF areas

An OSPF network can be divided into *areas* that are logical groupings of hosts and networks. An area includes its connecting router having interfaces connected to the network. Each area maintains a separate link state database whose information may be summarized towards the rest of the network by the connecting router. Thus, the topology of an area is unknown outside the area. This reduces the routing traffic between parts of an autonomous system.

Areas are uniquely identified with 32-bit numbers. The area identifiers are commonly written in the dot-decimal notation, familiar from IPv4 addressing. However, they are not IP addresses and may duplicate, without conflict, any IPv4 address. The area identifiers for IPv6 implementations (OSPFv3) also use 32-bit identifiers written in the same notation. When dotted formatting is omitted, most implementations expand area 1 to the area identifier 0.0.0.1, but some have been known to expand it as 1.0.0.0.

OSPF defines several special area types:

### Backbone area

The backbone area (also known as *area 0* or *area 0.0.0.0*) forms the core of an OSPF network. All other areas are connected to it, either directly or through other routers. Inter-area routing happens via routers connected to the backbone area and to their own associated areas. It is the logical and physical structure for the 'OSPF domain' and is attached to all nonzero areas in the OSPF domain. Note that in OSPF the term Autonomous System Boundary Router (ASBR) is historic, in the sense that many OSPF domains can coexist in the same Internet-visible autonomous system, RFC 1996.

The backbone area is responsible for distributing routing information between nonbackbone areas. The backbone must be contiguous, but it does not need to be physically contiguous; backbone connectivity can be established and maintained through the configuration of virtual links.

All OSPF areas must connect to the backbone area. This connection, however, can be through a virtual link. For example, assume area 0.0.0.1 has a physical connection to area 0.0.0.0. Further assume that area 0.0.0.2 has no direct connection to the backbone, but this area does have a connection to area 0.0.0.1. Area 0.0.0.2 can use a virtual link through the *transit area* 0.0.0.1 to reach the backbone. To be a transit area, an area has to have the transit attribute, so it cannot be stubby in any way.

### Stub area

A stub area is an area which does not receive route advertisements external to the AS and routing from within the area is based entirely on a default route. An ABR deletes type 4, 5 LSAs from internal routers, sends them a default route of 0.0.0.0 and turns itself into a default gateway. This reduces LSDB and routing table size for internal routers.

Modifications to the basic concept of stub areas exist in the *not-so-stubby* area (NSSA). In addition, several other proprietary variations have been implemented by systems vendors, such as the *totally stubby area* (TSA) and the *NSSA totally stubby area*, both an extension in Cisco Systems routing equipment.

## Not-so-stubby area

A *not-so-stubby area* (NSSA) is a type of stub area that can import autonomous system external routes and send them to other areas, but still cannot receive AS-external routes from other areas.<sup>[16]</sup> NSSA is an extension of the stub area feature that allows the injection of external routes in a limited fashion into the stub area. A case study simulates an NSSA getting around the Stub Area problem of not being able to import external addresses. It visualizes the following activities: the ASBR imports external addresses with a type 7 LSA, the ABR converts a type 7 LSA to type 5 and floods it to other areas, the ABR acts as an "ASBR" for other areas. The ASBRs do not take type 5 LSAs and then convert to type 7 LSAs for the area.

## Proprietary extensions

Several vendors (Cisco, Allied Telesis, Juniper, Alcatel-Lucent, Huawei, Quagga), implement the two extensions below for stub and not-so-stubby areas. Although not covered by RFC standards, they are considered by many to be standard features in OSPF implementations.

## Totally stubby area

A *totally stubby area* is similar to a stub area. However, this area does not allow *summary* routes in addition to not having *external* routes, that is, *inter-area* (IA) routes are not summarized into totally stubby areas. The only way for traffic to get routed outside the area is a default route which is the only Type-3 LSA advertised into the area. When there is only one route out of the area, fewer routing decisions have to be made by the route processor, which lowers system resource utilization.

Occasionally, it is said that a TSA can have only one ABR.

## NSSA totally stubby area

An addition to the standard functionality of an NSSA, the *totally stubby NSSA* is an NSSA that takes on the attributes of a TSA, meaning that type 3 and 4 summary routes are not flooded into this type of area. It is also possible to declare an area both totally stubby and not-so-stubby, which means that the area will receive only the default route from area 0.0.0.0, but can also contain an autonomous system boundary router (ASBR) that accepts external routing information and injects it into the local area, and from the local area into area 0.0.0.0.

Redistribution into an NSSA area creates a special type of LSA known as type 7, which can exist only in an NSSA area. An NSSA ASBR generates this LSA, and an NSSA ABR router translates it into type 5 LSA which gets propagated into the OSPF domain.

A newly acquired subsidiary is one example of where it might be suitable for an area to be simultaneously not-so-stubby and totally stubby if the practical place to put an ASBR is on the edge of a totally stubby area. In such a case, the ASBR does send externals into the totally stubby area, and they are available to OSPF speakers within that area. In Cisco's implementation, the external routes can be summarized before injecting them into the totally stubby area. In general, the ASBR should not advertise default into the TSA-NSSA, although this can work with extremely careful design and operation, for the limited special cases in which such an advertisement makes sense.

By declaring the totally stubby area as NSSA, no external routes from the backbone, except the default route, enter the area being discussed. The externals do reach area 0.0.0.0 via the TSA-NSSA, but no routes

other than the default route enter the TSA-NSSA. Routers in the TSA-NSSA send all traffic to the ABR, except to routes advertised by the ASBR.

### **Transit area**

A transit area is an area with two or more OSPF border routers and is used to pass network traffic from one adjacent area to another. The transit area does not originate this traffic and is not the destination of such traffic.

### **Router types**

OSPF defines the following overlapping categories of routers:

#### **Internal router (IR)**

*An internal router has all its interfaces belonging to the same area.*

#### **Area border router (ABR)**

*An area border router is a router that connects one or more areas to the main backbone network.* It is considered a member of all areas it is connected to. An ABR keeps multiple instances of the link-state database in memory, one for each area to which that router is connected.

#### **Backbone router (BR)**

*A backbone router has an interface to the backbone area.* Backbone routers may also be area routers, but do not have to be.

#### **Autonomous system boundary router (ASBR)**

*An autonomous system boundary router is a router that is connected by using more than one routing protocol and that exchanges routing information with routers autonomous systems.* ASBRs typically also run an exterior routing protocol (e.g., BGP), or use static routes, or both. An ASBR is used to distribute routes received from other, external ASs throughout its own autonomous system. An ASBR creates External LSAs for external addresses and floods them to all areas via ABR. Routers in other areas use ABRs as next hops to access external addresses. Then ABRs forward packets to the ASBR that announces the external addresses.

The router type is an attribute of an OSPF process. A given physical router may have one or more OSPF processes. For example, a router that is connected to more than one area, and which receives routes from a BGP process connected to another AS, is both an area border router and an autonomous system boundary router.

Each router has an identifier, customarily written in the dotted decimal format (e.g., 1.2.3.4) of an IP address. This identifier must be established in every OSPF instance. If not explicitly configured, the highest logical IP address will be duplicated as the router identifier. However, since the router identifier is not an IP address, it does not have to be a part of any routable subnet in the network, and often isn't to avoid confusion.

## Router attributes

In addition to the four router types, OSPF uses the terms *designated router (DR)* and *backup designated router (BDR)*, which are attributes of a router interface.

### Designated router

A *designated router (DR)* is the router interface elected among all routers on a particular multiaccess network segment, generally assumed to be broadcast multiaccess. Special techniques, often vendor-dependent, may be needed to support the DR function on non-broadcast multiaccess (NBMA) media. It is usually wise to configure the individual virtual circuits of a NBMA subnet as individual point-to-point lines; the techniques used are implementation-dependent.

### Backup designated router

A *backup designated router (BDR)* is a router that becomes the designated router if the current designated router has a problem or fails. The BDR is the OSPF router with second highest priority at the time of the last election.

A given router can have some interfaces that are designated (DR) and others that are backup designated (BDR), and others that are non-designated. If no router is a DR or a BDR on a given subnet, the BDR is first elected, and then a second election is held for the DR. The DR is elected based on the following default criteria:

- If the priority setting on an OSPF router is set to 0, that means it can NEVER become a DR or BDR.
- When a DR fails and the BDR takes over, there is another election to see who becomes the replacement BDR.
- The router sending the Hello packets with the highest priority wins the election.
- If two or more routers tie with the highest priority setting, the router sending the Hello with the highest RID (Router ID) wins. NOTE: a RID is the highest logical (loopback) IP address configured on a router, if no logical/loopback IP address is set then the router uses the highest IP address configured on its active interfaces (e.g. 192.168.0.1 would be higher than 10.1.1.2).
- Usually the router with the second highest priority number becomes the BDR.
- The priority values range between 0 – 255, with a higher value increasing its chances of becoming DR or BDR.
- If a higher priority OSPF router comes online after the election has taken place, it will not become DR or BDR until (at least) the DR and BDR fail.
- If the current DR 'goes down' the current BDR becomes the new DR and a new election takes place to find another BDR. If the new DR then 'goes down' and the original DR is now available, still previously chosen BDR will become DR.

DR's exist for the purpose of reducing network traffic by providing a source for routing updates. The DR maintains a complete topology table of the network and sends the updates to the other routers via multicast. All routers in a multi-access network segment will form a slave/master relationship with the

DR. They will form adjacencies with the DR and BDR only. Every time a router sends an update, it sends it to the DR and BDR on the multicast address 224.0.0.6. The DR will then send the update out to all other routers in the area, to the multicast address 224.0.0.5. This way all the routers do not have to constantly update each other, and can rather get all their updates from a single source. The use of multicasting further reduces the network load. DRs and BDRs are always setup/elected on OSPF broadcast networks. DR's can also be elected on NBMA (Non-Broadcast Multi-Access) networks such as Frame Relay or ATM. DRs or BDRs are not elected on point-to-point links (such as a point-to-point WAN connection) because the two routers on either sides of the link must become fully adjacent and the bandwidth between them cannot be further optimized. DR and non-DR routers evolve from 2-way to full adjacency relationships by exchanging DD, Request, and Update.